

Topics Covered:

- **Cryptography**



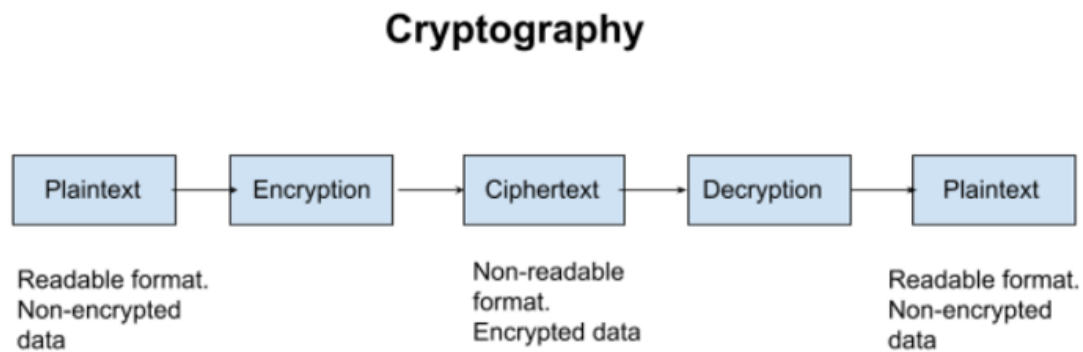
Description:

In this lab, you will learn the basics of cryptography, algorithm description and its implementation in Python.

Terminologies of Cryptography:

The frequently used terms in cryptography are explained here:

1. Plain Text: The plain text message is the text which is readable and can be understood by all users. The plain text is the message which undergoes cryptography.
2. Cipher Text: Cipher text is the message obtained after applying cryptography on plain text.
3. Encryption: The process of converting plain text to cipher text is called encryption. It is also called as encoding.
4. Decryption: The process of converting cipher text to plain text is called decryption. It is also termed as decoding.



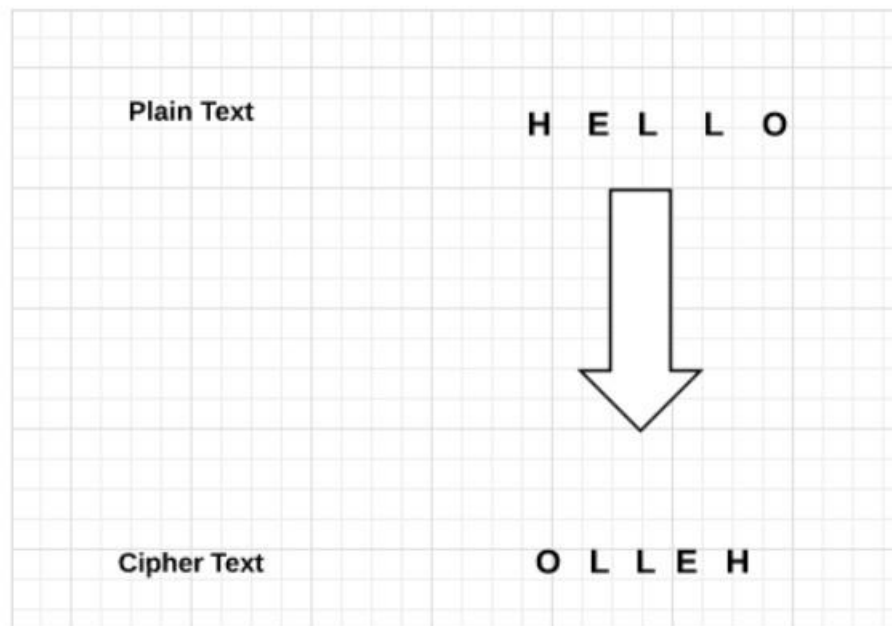
1. Cryptography with python- Reverse Cipher

The algorithm of reverse cipher holds the following features:

- Reverse Cipher uses a pattern of reversing the string of plain text to convert as cipher text.
- The process of encryption and decryption is the same.
- To decrypt cipher text, the user simply needs to reverse the cipher text to get the plain text.

Drawback The major drawback of the reverse cipher is that it is very weak. A hacker can easily break the cipher text to get the original message. Hence, reverse cipher is not considered as good option to maintain secure communication channel.





Example

Consider an example where the statement “This is program to explain reverse cipher is to be implemented with reverse cipher algorithm “.

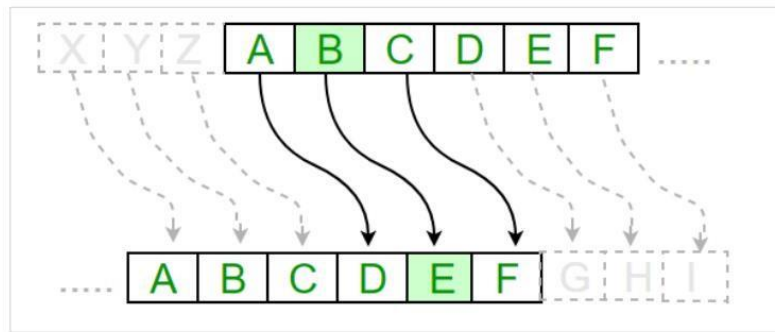
2. Cryptography with python- Caesar Cipher

The algorithm of Caesar cipher holds the following features:

- Caesar Cipher Technique is the simple and easy method of encryption technique.
- It is simple type of substitution cipher.
- Each letter of plain text is replaced by a letter with some fixed number of positions down with alphabet.



The following diagram depicts the working of Caesar cipher algorithm implementation:



Example

Consider an example where the statement “ CEASER CIPHER DEMO “ and with Key = 4 to be implemented using Caesar Cipher.

Hacking of Caesar Cipher Algorithm

The cipher text can be hacked with various possibilities. One of such possibilities is Brute Force Technique, which involves trying every possible decryption key. This technique does not demand much effort and is relatively simple for a hacker.

3. Cryptography with python- Monoalphabetic Cipher

A Monoalphabetic cipher uses a fixed substitution for encrypting the entire message. A monoalphabetic cipher using a Python dictionary is shown here:

```
monoalpha_cipher = {  
    'a': 'm', 'b': 'n', 'c': 'b', 'd': 'v', 'e': 'c', 'f': 'x', 'g': 'z', 'h': 'a', 'i': 's', 'j': 'd', 'k': 'f', 'l': 'g', 'm': 'h',  
    'n': 'j', 'o': 'k', 'p': 'l', 'q': 'p', 'r': 'o', 's': 'i', 't': 'u', 'u': 'y', 'v': 't', 'w': 'r', 'x': 'e', 'y': 'w', 'z': 'q', ' ': ' ',  
}
```

Example

Consider an example where the statement “ defend the east wall of the castle “ to be implemented using Monoalphabetic Cipher.

